Biocontrol modeling project: Environment Setup and Installation Manual

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1 Introduction

This guide provides a step-by-step visual tutorial to help you set up your development environment and run the project locally. Includes screenshots and detailed instructions for each step.

By the end of this guide, you will have:

- Installed Python and Visual Studio Code.
- Cloned the project repository directly from Visual Studio Code.
- Created and activated a virtual environment.
- Installed all the required Python libraries.

Make sure to follow each step carefully and verify that the output matches the screenshots provided.

2 Python Installation

Python is the programming language used in this project. In this guide, we will use Python version 3.10.9, which is required for compatibility with the project dependencies.

Here is how to install it correctly on your computer.

2.1 Step 1: Access to the official download page

Go to the official Python page using the following link:

`https://www.python.org/downloads/windows/`

2.2 Step 2: Download the installer

And look for Python version 3.10.9 (64-bit)

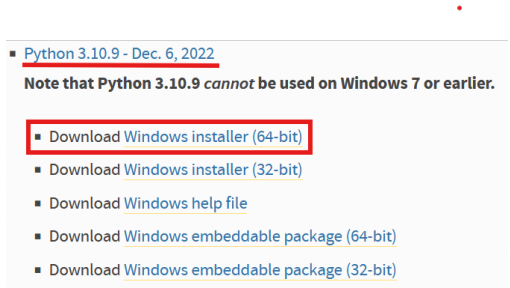

Figure 1: Python download official page.

2.3 Step 3: Execute the installer

When the download file is executed, the following window will appear.

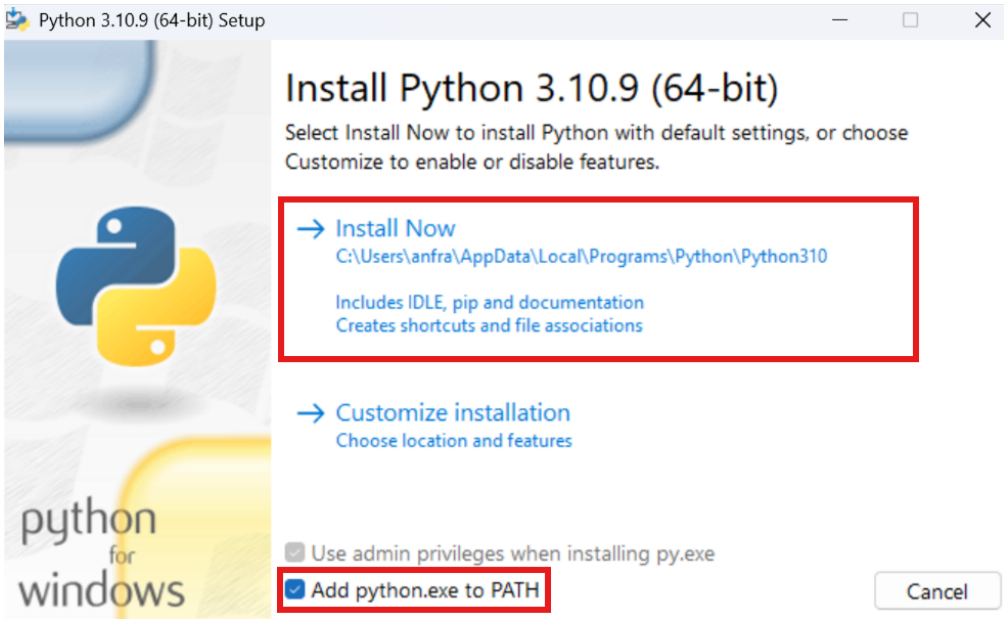

Figure 2: Python installer window.

1. Check the box that says **"Add python.exe to PATH"** (is essential for it to work from the terminal).
2. Then click on **"Install Now"**to start the installation.

2.4 Step 4: Check the installation

Once the installation is complete, open the terminal (or cmd) and type the following command. You can type windows key + R to open the terminal.

```
python --version
```

If everything is correct, you will see the following message:

```
Python 3.10.9
```

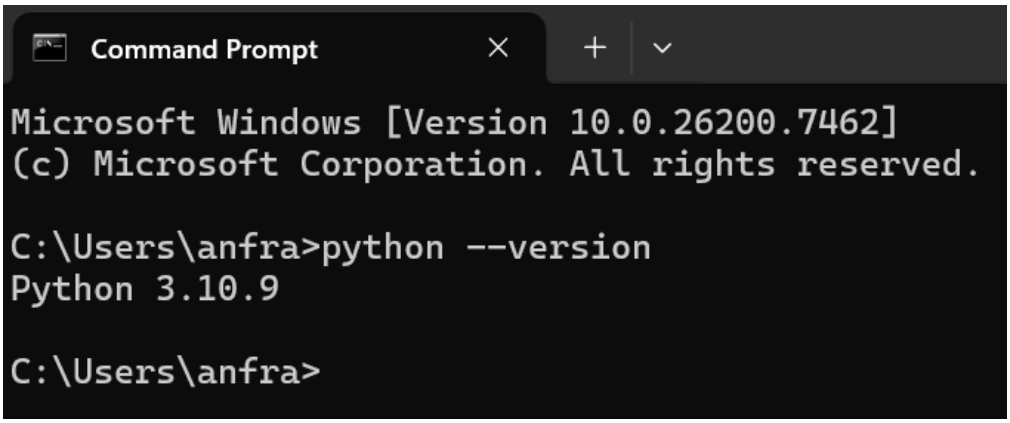

Figure 3: Command prompt showing the installed Python version.

If a different version appears, you may have multiple Python versions installed. In that case, make sure to use Python 3.10 when creating the virtual environment.(See 5)

3 Installing Visual Studio Code

Visual Studio Code (VS Code) is a lightweight and powerful code editor that supports Python development through extensions. In this section, you will learn how to download and install VS Code on your system.

3.1 Step 1: Access to the official website

Go to the official Visual Studio Code website using the following link:

`https://code.visualstudio.com/`

3.2 Step 2: Download the installer

The homepage should look like the following image:

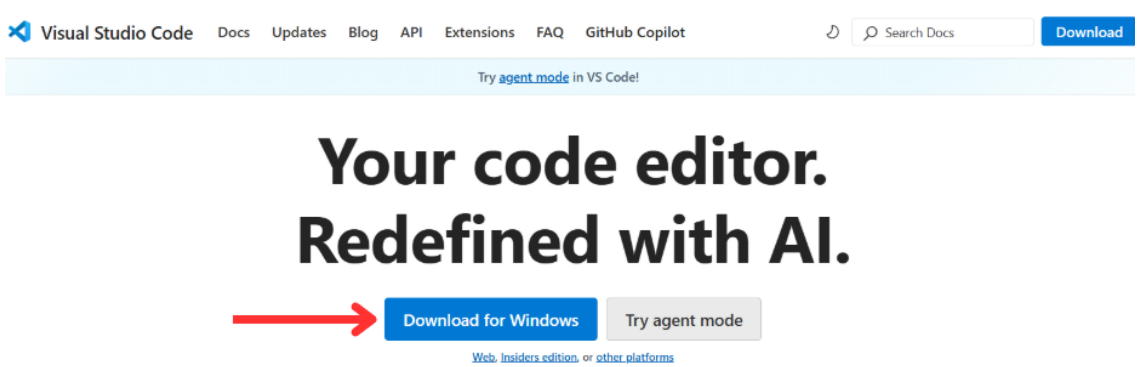

Figure 4: VS Code download page for Windows.

1. Click on the **"Download for Windows"** button to download the installer.

3.3 Step 3: Run the installer

After downloading, open the installer file. The setup wizard will appear:

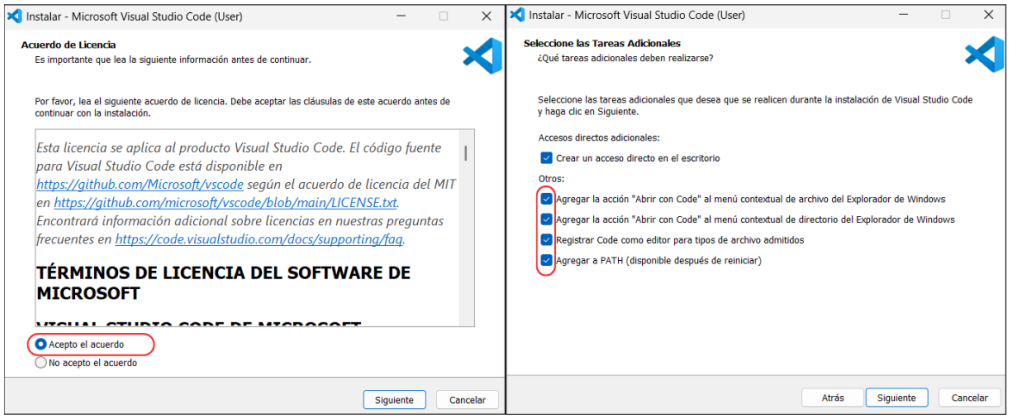

Figure 5: VS Code installation setup window.

1. Accept the license agreement and click **"Next"**.
2. Choose the installation folder or leave the default, then click **"Next"**.
3. In the "Select Additional Tasks" step, make sure to check:
   - **"Add to PATH"**
   - **"Register Code as an editor for supported file types"**
   - **"Add 'Open with Code' action to Windows Explorer"**
4. Click **"Install"** to begin the installation.

3.4 Step 4: Launch VS Code

Once the installation is complete, you can open Visual Studio Code from the Start Menu or by typing:

```
code
```

in the command prompt (only if you added it to the PATH during setup).

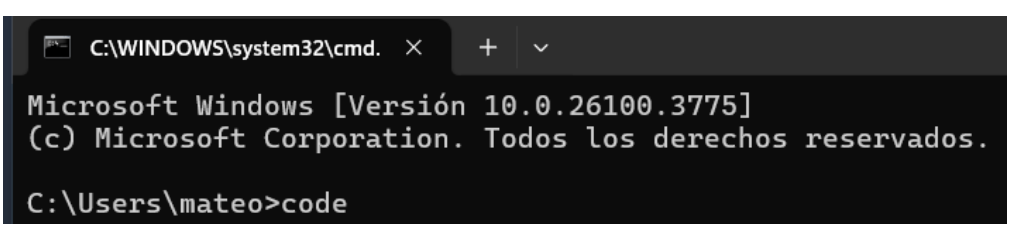

Figure 6: Opening VS Code from the terminal using `code`.

4 Cloning the GitHub Repository

Before you can clone the project repository, you need to have:

- Git installed on your system (see previous section).
- A GitHub account already created.
- You must be signed in to GitHub within Visual Studio Code.

4.1 Step 1: Sign in to GitHub in VS Code

Open Visual Studio Code. On the bottom-left corner, click on the account icon and select **"Sign in to GitHub"** or **"Sign in to GitHub to use GitHub copilot"**.

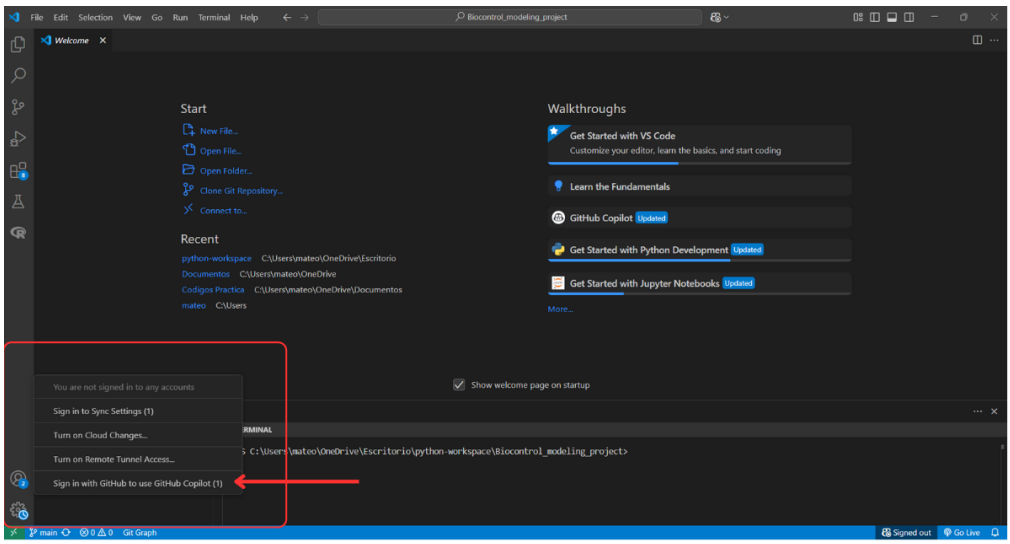

Figure 7: Signing in to GitHub in VS Code.

Follow the prompts to complete the login process.

4.2 Step 2: Clone the repository

- Click on file section, then click on new window

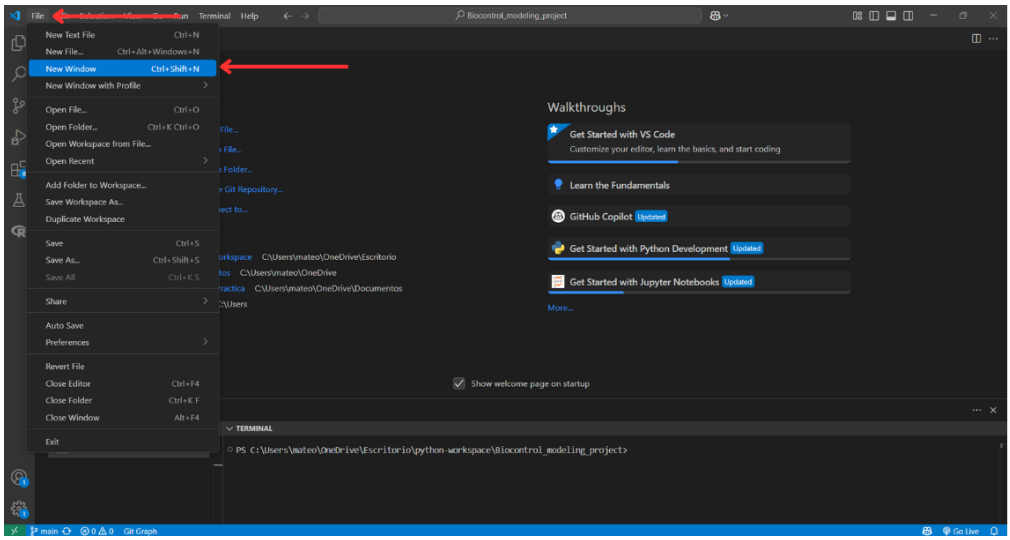

Figure 8: Accessing the new window option in VS Code.

- Click on **Clone Git Repository...**

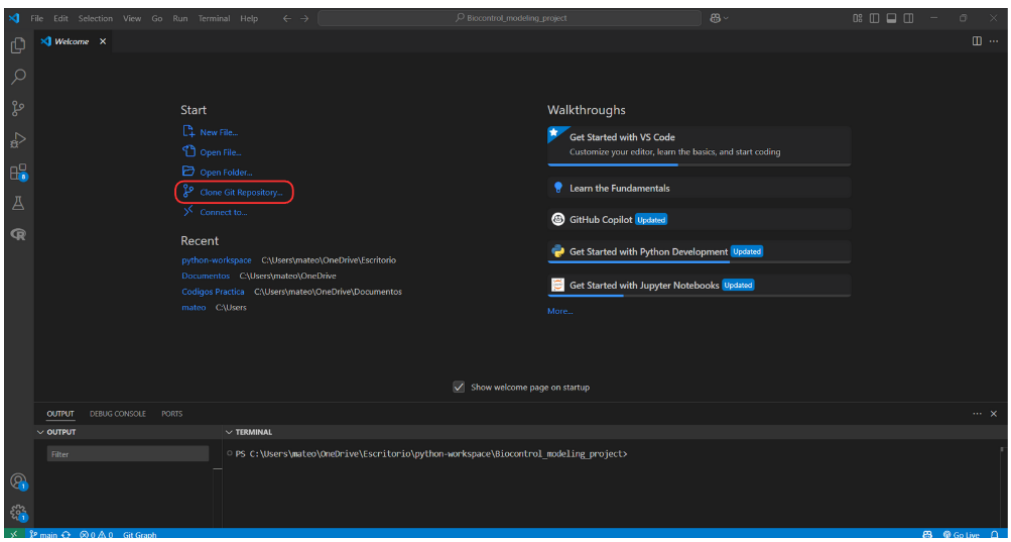

Figure 9: Accessing to Clone Git Repository window.

- Paste the repository URL in the input box that appears at the top.

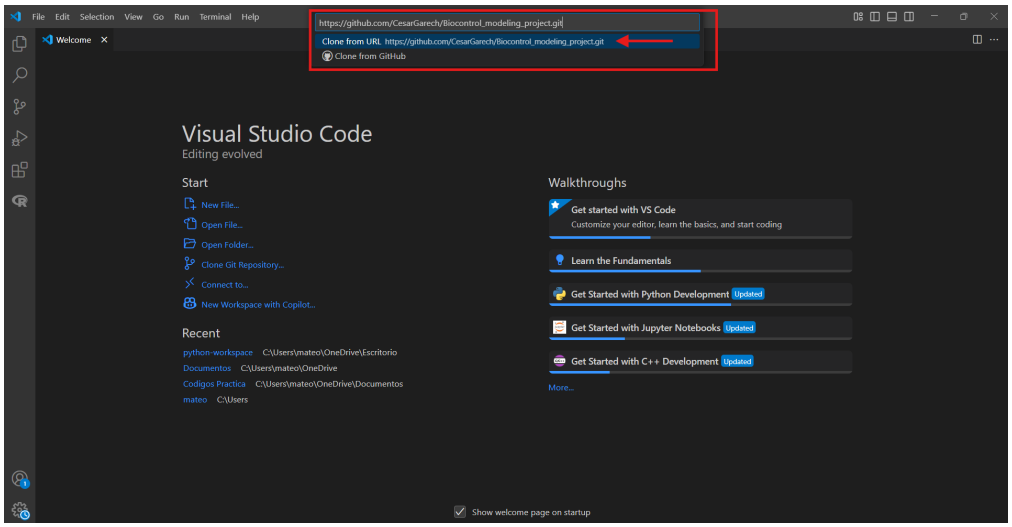

Figure 10: Copy the url repository.

- Choose a folder on your local machine where the repository will be cloned.

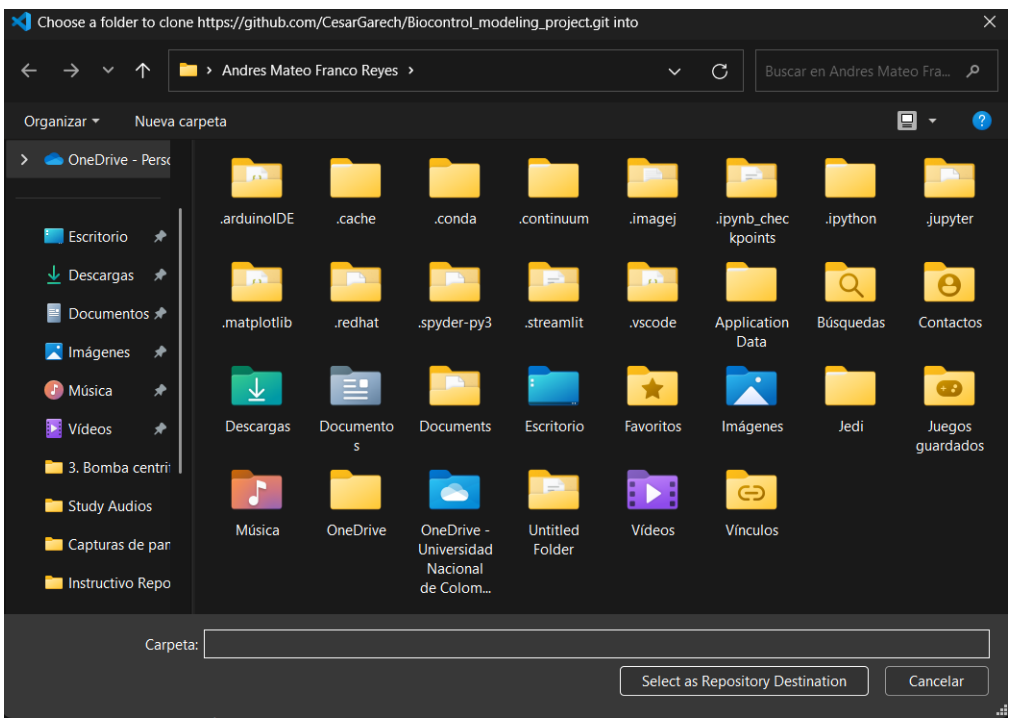

Figure 11: Choose a folder to cloned the repository.

4.3 Step 3: Confirm the project was cloned

Once you finish to choose a destination folder, you should see the project structure on the left panel (Explorer view). You can now start working with the code

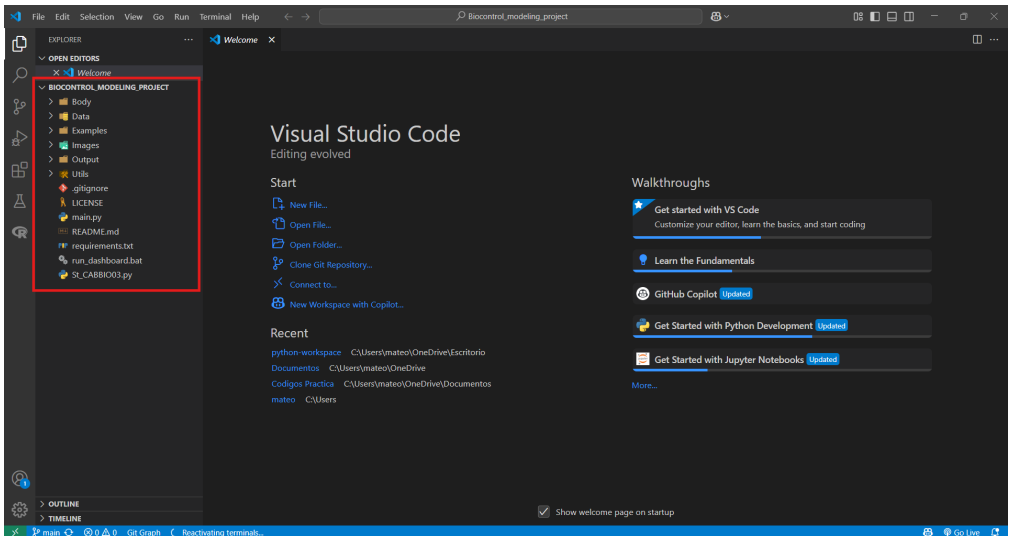

Figure 12: Repository cloned.

5 Creating and Activating the Virtual Environment

To isolate the project dependencies and avoid conflicts with global packages, it is recommended to use a virtual environment.

5.1 Step 1: Open the built-in terminal in VS Code

With the project opened in Visual Studio Code, go to **Terminal > New Terminal** to open the built-in terminal.

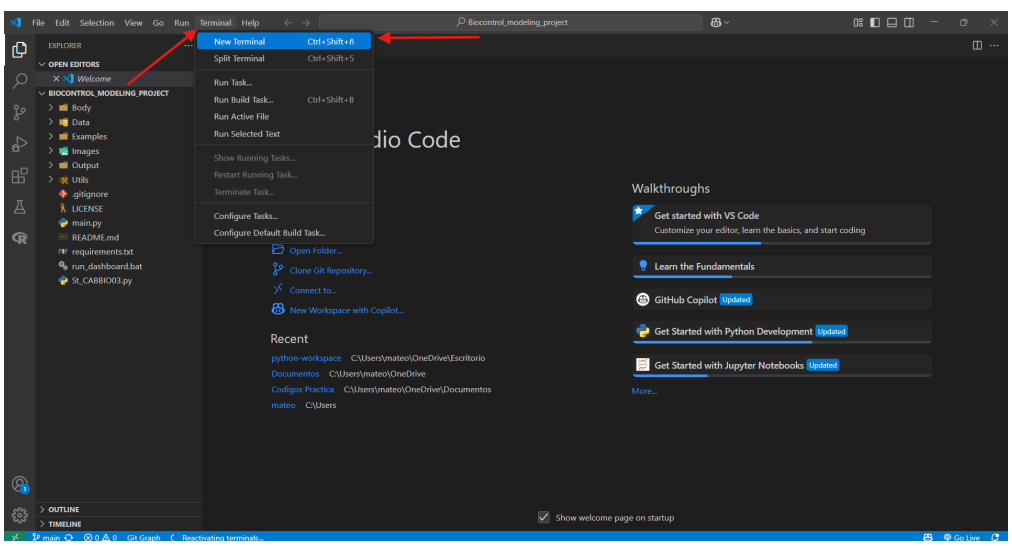

Figure 13: Opening the integrated terminal in VS Code.

5.2 Step 2: Create the virtual environment

If you have multiple Python versions installed, it is important to explicitly use Python 3.10 to create the virtual environment.

In the terminal, type the following command and press `Enter`:

```
py -3.10 -m venv venv
```

This will create a folder named 'venv' containing the virtual environment using Python 3.10.

After creating the environment, you can verify the Python version with:

```
venv\Scripts\python --version
```

You must see Python 3.10.9.

5.3 Step 3: Activate the virtual environment

Depending on your operating system, use the appropriate command:

On Windows:

```
venv\Scripts\activate
```

On macOS / Linux:

```
source venv/bin/activate
```

Once activated, you will see the environment name (‘venv‘) in parentheses at the beginning of the terminal line.

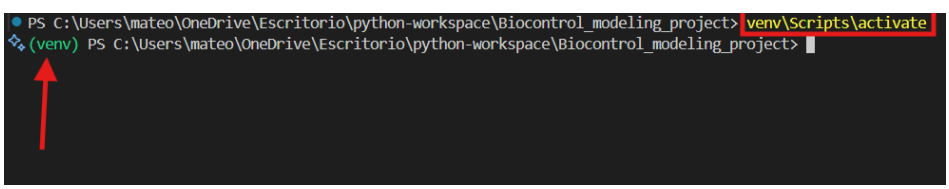

Figure 14: Virtual environment activated in the terminal.

5.4 Step 4: Select the Python interpreter in VS Code

To ensure Visual Studio Code uses the correct Python version:

1. Press Ctrl + Shift + P
2. Type: Python: Select Interpreter
3. Select the interpreter located in:

```
venv\Scripts\python.exe
```

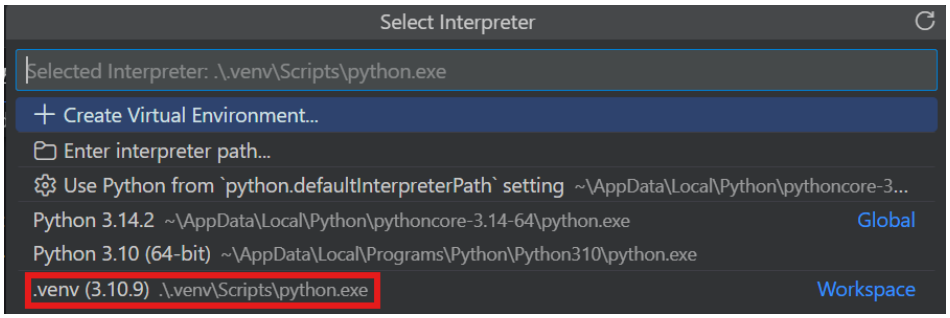

Figure 15: Selecting interpreter

5.5 Step 5: Install the required dependencies

In your project includes a `requirements.txt` file, run the following command to install the required libraries:

```
pip install -r requirements.txt
```

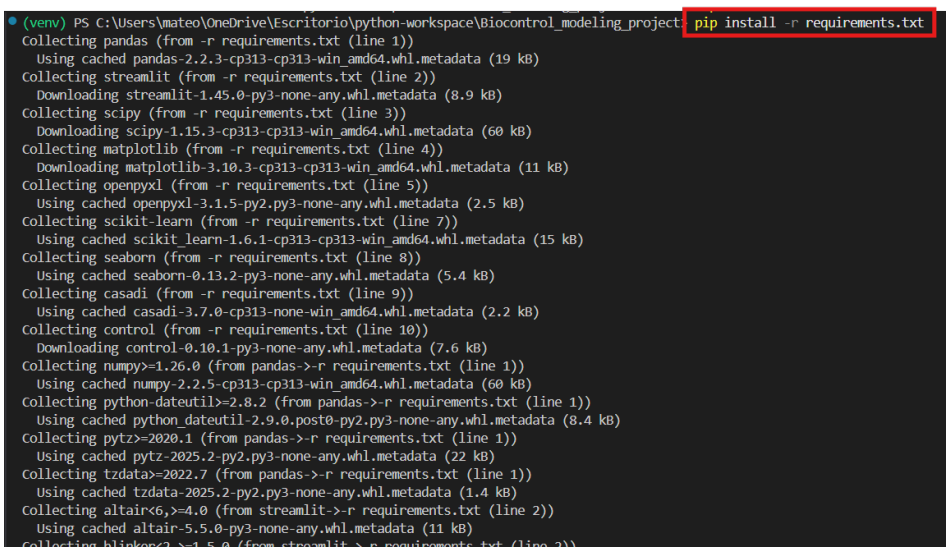

Figure 16: Installing the dependencies listed in `requirements.txt`.